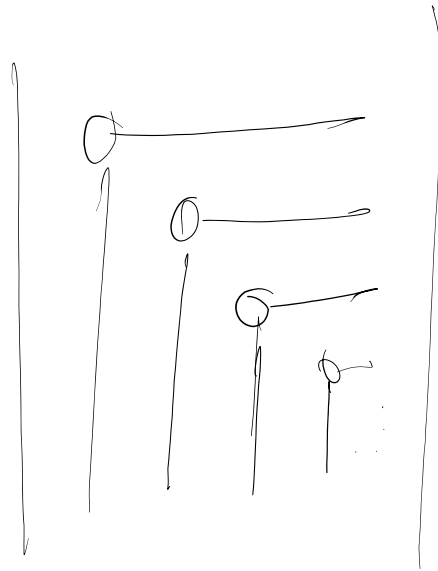


Cost of Elimination

For $n \times n$ matrix,

the number of operations required
in eliminating the first column

is n^2 . For the next column, it is $(n-1)^2$



$$\text{So, } 1^2 + 2^2 + 3^2 + \dots + (n-1)^2 + n^2$$

$$= \frac{n(n+1)(2n+1)}{6} \approx n^3 \rightarrow O(n^3)$$

(considering worst case)

Factorization of $A = LU$

$\rightarrow$ Upper Triangular Matrix.
 $\rightarrow$ Lower Triangular Matrix

To solve $Ax=b$, we used $E \cdot A = U \rightarrow$ Upper Triangular Matrix.

$\rightarrow E_{32} E_{31} E_{21}$ $\hookrightarrow$ Also can be called REF.

$\rightarrow$ These are derived
from Gaussian Elimination.

Example:

$$A = \begin{bmatrix} 2 & 1 \\ 8 & 7 \end{bmatrix}$$

Now,

$$E_{21} A = \begin{bmatrix} 1 & 0 \\ -4 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 8 & 7 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} = U$$

$$\hookrightarrow A = E_{21}^{-1} U$$

$$= L U \quad [E_{21}^{-1} = L \text{ and here } 1 \text{ has to be pivot}]$$

Now,

$$A = \begin{bmatrix} 1 & 0 \\ -4 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1/2 \\ 0 & 1 \end{bmatrix}$$

L

D

U $\rightarrow$ Transformed the pivot in 1

$\hookrightarrow$ Diagonal

Example:

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 2 & 2 \\ 3 & 4 & 5 \end{bmatrix}$$

$$\hookrightarrow E_{31} A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 4 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 2 & 2 & 2 \\ 3 & 4 & 5 \end{bmatrix}$$

$$\hookrightarrow E_{31} E_{21} A = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 2 & 2 & 2 \\ 0 & 4 & 2 \end{bmatrix}$$

$$\hookrightarrow E_{32} E_{31} E_{21} A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 4 & 2 \end{bmatrix}$$

$$\hookrightarrow EA = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} = U$$

$$\therefore E = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -3 & -2 & 1 \end{bmatrix} \Rightarrow E^{-1} = L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix}$$

$$\text{Now, } U = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

page 32

$$\Rightarrow U = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$\therefore A = L U = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

From Gilbert Strang Page 3

Example:



Can't write the
same stuff again